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# ============================================================
# DESCRIPTIVE ANALYSIS - CREDIT CARD APPROVAL PREDICTION
# (Without Uploading CSV File)
# ============================================================

# Step 1: Import Libraries
import pandas as pd
import numpy as np

# ============================================================
# Step 2: Create Sample Dataset
# ============================================================

data = {
    'ID':[1,2,3,4,5,6,7,8,9,10],
    'CODE_GENDER':['M','F','F','M','M','F','M','F','M','F'],
    'FLAG_OWN_CAR':['Y','N','Y','Y','N','N','Y','N','Y','N'],
    'FLAG_OWN_REALTY':['Y','Y','N','Y','Y','N','Y','N','Y','Y'],
    'CNT_CHILDREN':[0,1,0,2,1,0,3,1,0,2],
    'AMT_INCOME_TOTAL':[180000,120000,250000,150000,200000,170000,300000,110000,220000,190000],
    'DAYS_BIRTH':[-12000,-15000,-17000,-14000,-18000,-13000,-16000,-19000,-12500,-17500],
    'CNT_FAM_MEMBERS':[2,3,2,4,3,2,5,3,2,4]
}

app = pd.DataFrame(data)

# ============================================================
# Step 3: Display Dataset
# ============================================================

print("First Five Rows")
print(app.head())

print("\nLast Five Rows")
print(app.tail())

# ============================================================
# Step 4: Shape
# ============================================================

print("\nShape of Dataset")
print(app.shape)

# ============================================================
# Step 5: Columns
# ============================================================

print("\nColumn Names")
print(app.columns)

# ============================================================
# Step 6: Data Types
# ============================================================

print("\nData Types")
print(app.dtypes)

# ============================================================
# Step 7: Dataset Information
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# ==========================================================

print("\nDataset Information")
app.info()

# ==========================================================
# Step 8: Missing Values
# ==========================================================

print("\nMissing Values")
print(app.isnull().sum())

# ==========================================================
# Step 9: Duplicate Rows
# ==========================================================

print("\nDuplicate Rows")
print(app.duplicated().sum())

# ==========================================================
# Step 10: Descriptive Statistics
# ==========================================================

print("\nDescriptive Statistics")
print(app.describe(include='all'))

# ==========================================================
# Step 11: Mean
# ==========================================================

print("\nMean")
print(app.mean(numeric_only=True))

# ==========================================================
# Step 12: Median
# ==========================================================

print("\nMedian")
print(app.median(numeric_only=True))

# ==========================================================
# Step 13: Mode
# ==========================================================

print("\nMode")
print(app.mode().iloc[0])

# ==========================================================
# Step 14: Standard Deviation
# ==========================================================

print("\nStandard Deviation")
print(app.std(numeric_only=True))

# ==========================================================
# Step 15: Variance
# ==========================================================

print("\nVariance")
print(app.var(numeric_only=True))

#
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# ==========================================================
# Step 16: Minimum Values
# ==========================================================

print("\nMinimum Values")
print(app.min(numeric_only=True))

# ==========================================================
# Step 17: Maximum Values
# ==========================================================

print("\nMaximum Values")
print(app.max(numeric_only=True))

# ==========================================================
# Step 18: Quartiles
# ==========================================================

print("\nQuartiles")
print(app.quantile([0.25,0.50,0.75], numeric_only=True))

# ==========================================================
# Step 19: Save Statistics
# ==========================================================

app.describe(include='all').to_csv("Descriptive_Statistics.csv")

print("\nDescriptive Statistics saved successfully.")

print("\n==============================")
print("DESCRIPTIVE ANALYSIS COMPLETED")
print("==============================")
```

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Missing Values
ID                  0
CODE_GENDER         0
FLAG_OWN_CAR        0
FLAG_OWN_REALTY     0
CNT_CHILDREN        0
AMT_INCOME_TOTAL    0
DAYS_BIRTH          0
CNT_FAM_MEMBERS     0
dtype: int64

Duplicate Rows
0

Descriptive Statistics
              ID CODE_GENDER FLAG_OWN_CAR FLAG_OWN_REALTY  CNT_CHILDREN  \
count   10.00000          10           10              10     10.000000
unique       NaN           2            2               2           NaN
top          NaN           M            Y               Y           NaN
freq         NaN           5            5               7           NaN
mean     5.50000         NaN          NaN             NaN      1.000000
std      3.02765         NaN          NaN             NaN      1.054093
min      1.00000         NaN          NaN             NaN      0.000000
25%      3.25000         NaN          NaN             NaN      0.000000
50%      5.50000         NaN          NaN             NaN      1.000000
75%      7.75000         NaN          NaN             NaN      1.750000
```